01 · PROBLEM & SYSTEM

Coal Rake Cycle

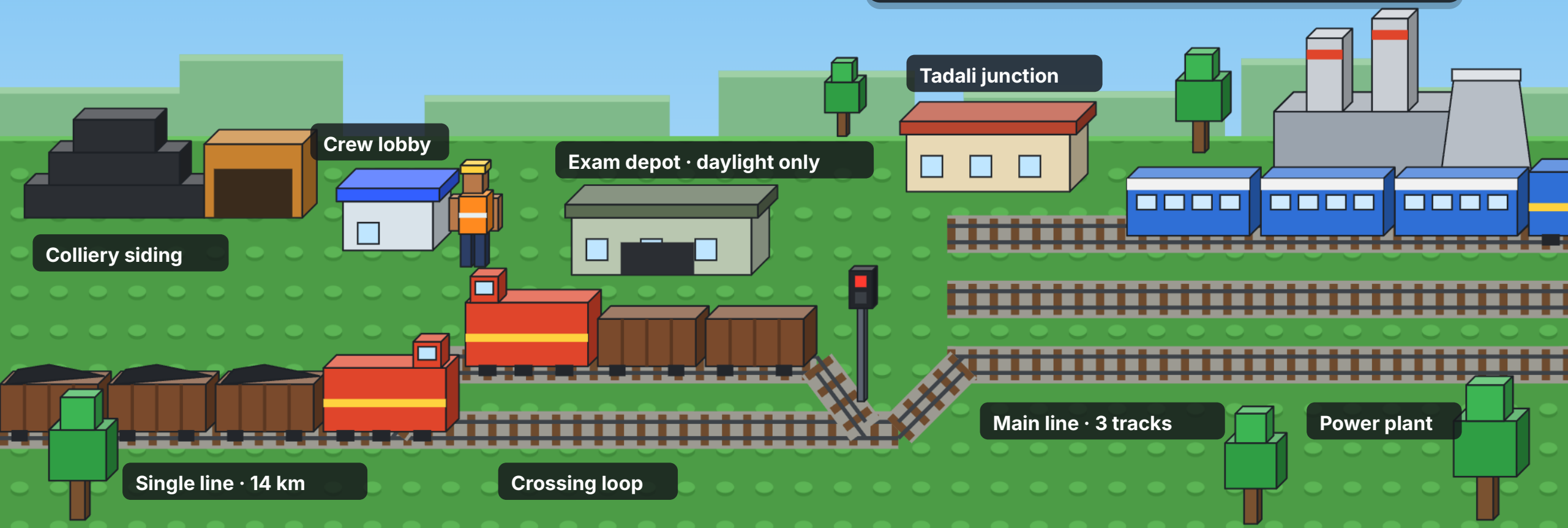
Ghugus → Tadali → Chandrapur · Central Railway

58-wagon BOXN rakes | 14 km single line | Closed loop

The problem

A rake's round trip takes far longer than its running time. Where does the extra time go?

1. Single-line crossings
2. **Waiting for a gap to merge at Tadali**
3. Expiring clocks: crew 9 h, BPC



02 · DATA

Where the numbers come from

No record? Expert min / likely / max → Triangular

Central Railway records

Loading time at siding
10–12 rakes · Gamma / Lognormal

Block running times
~30 trips each way

Other trains on single line
1 month of counts · Poisson

Main-line run, unloading
30 trips each · Lognormal

Public running status

Passenger times at Tadali
Timetable + daily lateness

30 days of history
Empirical lateness distribution

Refreshable daily
Digital-twin link for next-day paths

Staff, sheds & rules

Crew waiting
~50 events · crew lobby

Loco failures
Shed logs · Exp MTBF

Exam time
~100 man-hours / rake

Rules
9 h duty (+2) · BPC 6000 km / 30 d



Data access via Prof. Narayan Rangaraj

03 · QUESTIONS & OUTPUTS

What we'll test, what we'll measure

Goal: min E[cycle time] with rakes/day ≥ target · 30 days × 30 reps · CRN · 95% CI

What-if scenarios

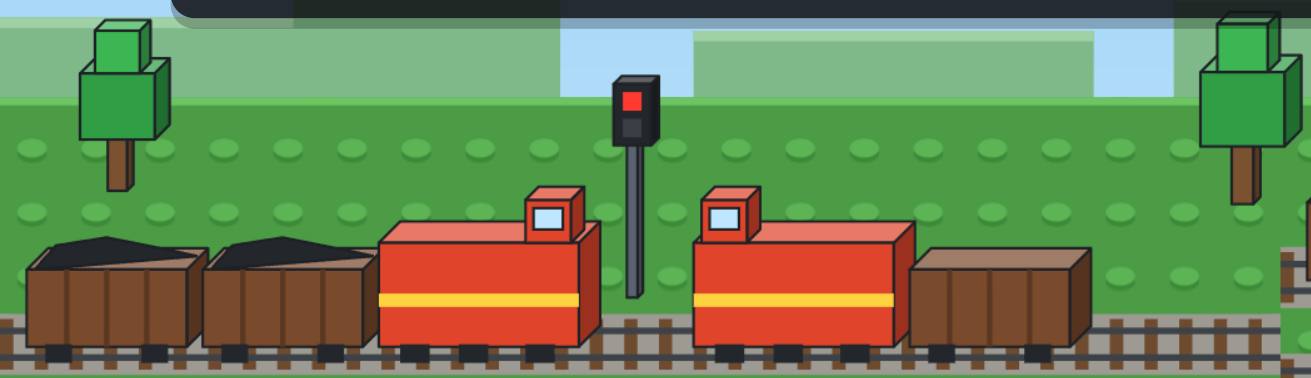
1. Base case — today's practice
2. **Dispatch** — batched vs alternating
3. **Tadali slots** — reserved paths vs take gaps
4. **Crew booking** — at loading vs when ready
5. **Exam trigger** — 3 / 5 / 7 days left vs expiry
6. **Fleet size** — 8 to 20 rakes
7. **Line doubling** — best policy vs ₹228 cr track

KPIs

Rake cycle time · mean & P90

split by cause: load, crew, single line, run, merge, relief, unload, exam

- Rakes/day & tonnes/day
- **Single-line utilization & wait**
- **Queue length at Tadali merge**
- Crew expiries per 100 trips
- Exam detention, h/rake/month



Single line: one rake waits



Doubled: both keep moving